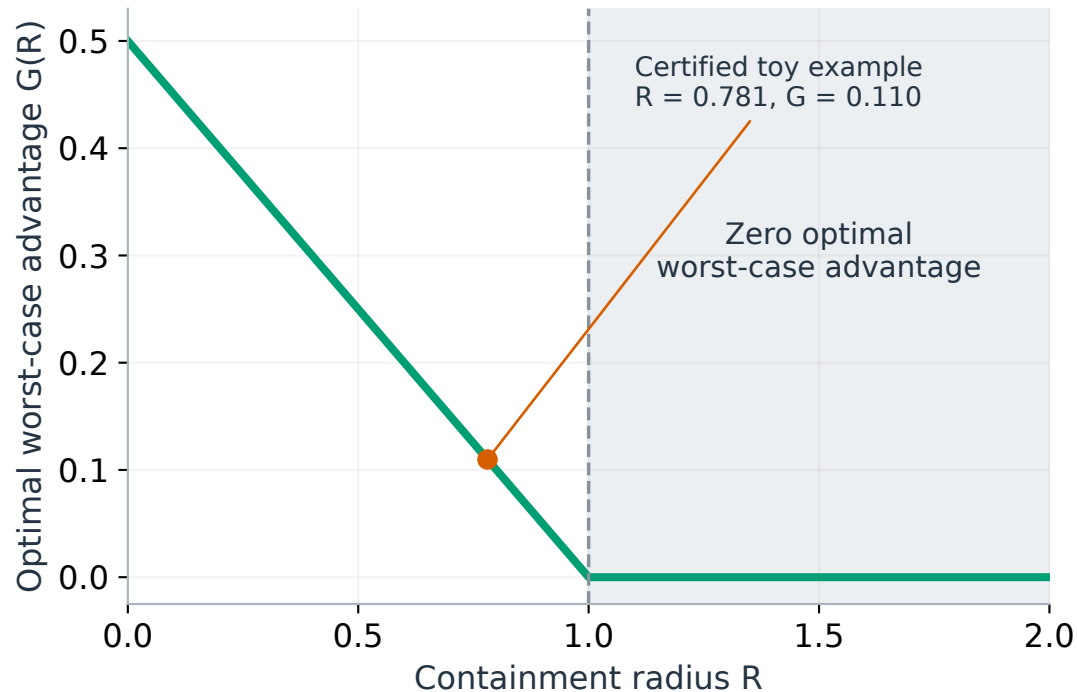


### 3 Prescribe relative to a reference



Curve: exact 16-action toy with a uniform reference;  $G(R) = \max\{0, (1 - R)/2\}$ . Not a general performance curve.

#### Containment $\rightarrow$ reference-relative safety

$$\Pr[V_{\theta^\circ}(\mu_R) \geq V_{\theta^\circ}(\mu_{\text{ref}})] \geq \gamma$$

With exchangeability and a valid width certificate.

#### For this translated span-ball prescription:

$R < 1$ : necessary, not sufficient, for positive advantage.

$R \geq 1$  gives  $G(R) = 0$ ; the reference is optimal.

This is not an impossibility for other reward sets.